

The background features a light green and yellow gradient with faint, stylized circuit board patterns. Various electronic components are scattered throughout, including microcontrollers, sensors, and connectors. In the top right corner, there is a detailed illustration of a blue RFID card with a colorful, abstract design and the word 'CARD' at the bottom. A rainbow-colored signal wave emanates from the top right of the card. In the bottom left corner, there are two wavy lines, one purple and one white, and a stylized keyboard icon. In the bottom right corner, there is a purple abstract shape.

Arduino — RFID

Access Control System

By : team Arduino

What is Arduino?

Meet Your Tiny Programmable Computer!

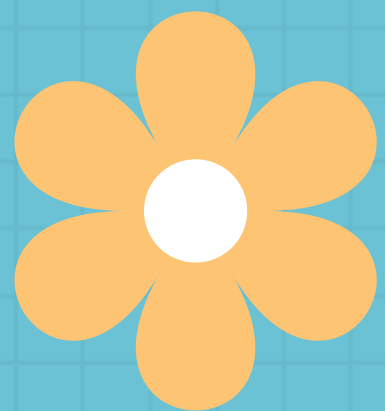
Arduino is a tiny computer that you can program to control things like lights, buzzers, and motors! It's like giving instructions to a robot brain - you tell it what to do, and it follows your commands to make cool projects come to life.

What is RFID?

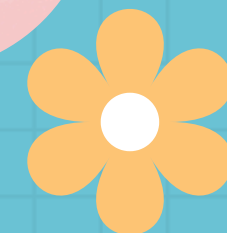
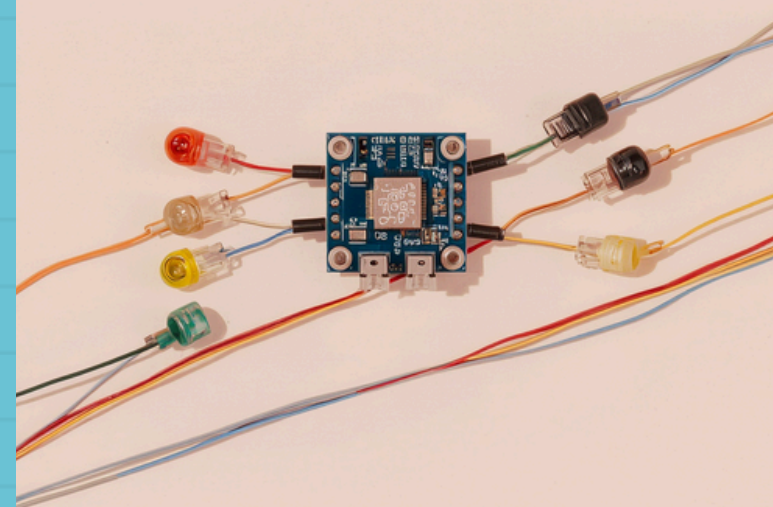
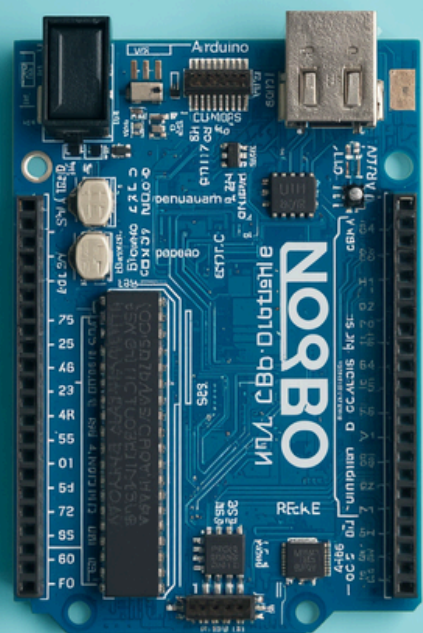
Understanding Radio Frequency Identification

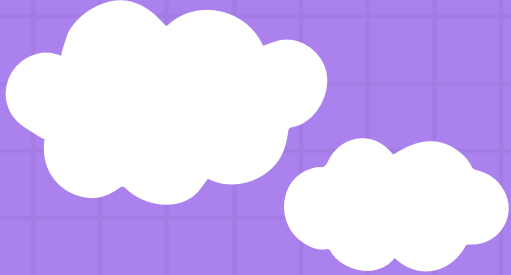
RFID means using special cards to unlock doors without keys! The card has a tiny chip inside that sends a secret code to the reader.

When you tap your RFID card near a reader, it instantly recognizes you – just like magic, but it's actually science!



Key Components for Our Project





How Does RFID Access Control Work?

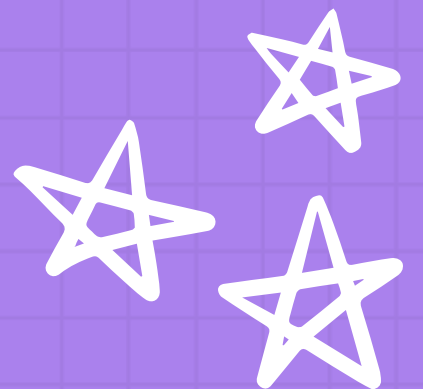


Simple Step-by-Step Process

When you scan your RFID card, the reader checks if it's allowed.

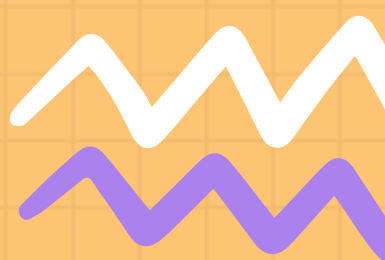
If yes, the door unlocks (LED lights up and buzzer beeps happily!).

If no, it stays locked and nothing happens.





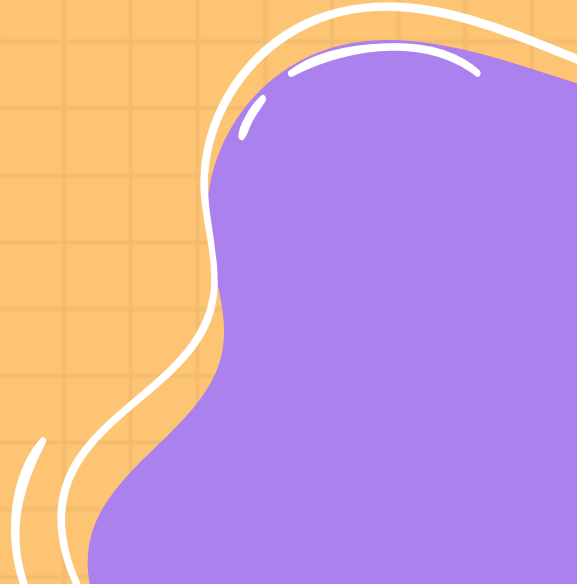
Before and After: Access Control in Action



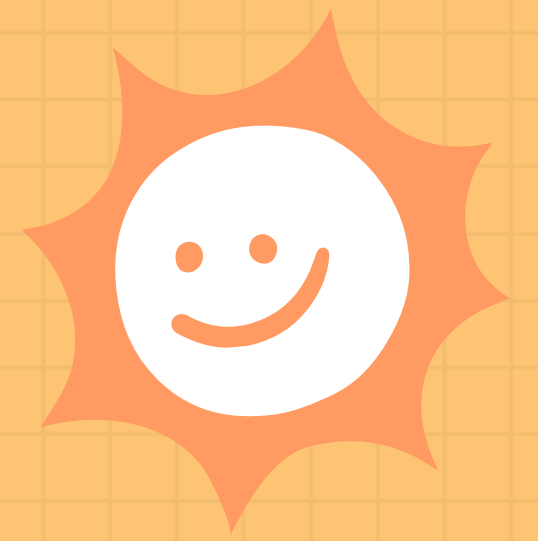
See How Your RFID System Works!

Before: Door is locked, red LED is off, buzzer stays silent – no access granted yet!

After: Door unlocks, green LED lights up bright, buzzer beeps happily – you're in!



Step 1 — Connect the Hardware

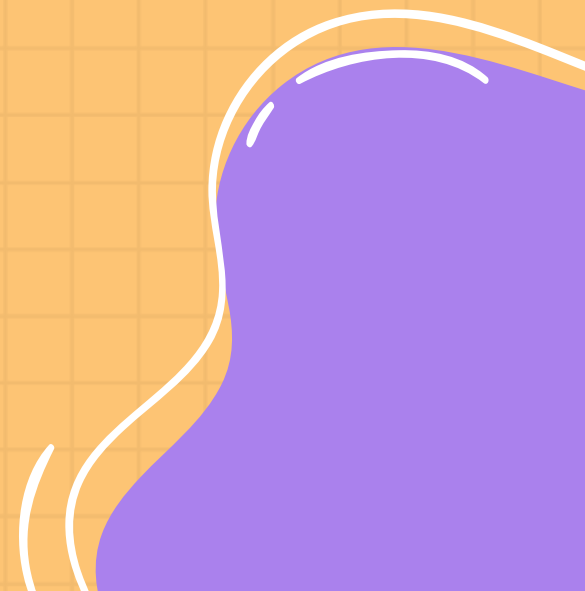
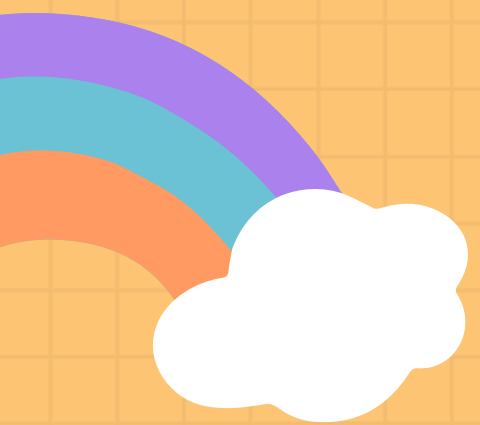


Building Your RFID Access Control Circuit

Attach the RFID reader module to the Arduino pins carefully following the pin diagram.

Connect the buzzer and LED lights to the correct Arduino pins for output signals.

Use jumper wires and breadboard to make neat, secure connections.





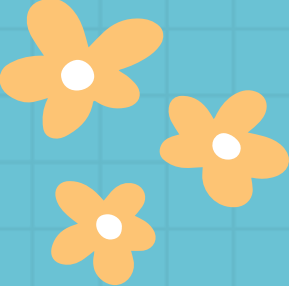
Wiring Diagram

Arduino Pin Connections

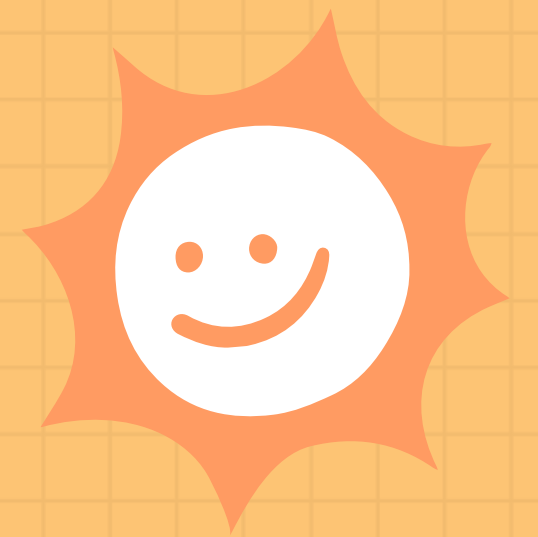
RFID Reader: Connect SDA to Pin 10, SCK to Pin 13, MOSI to Pin 11, MISO to Pin 12, RST to Pin 9.

Buzzer: Connect positive leg to Pin 8, negative to GND.

LEDs: Green LED to Pin 7, Red LED to Pin 6, both with resistors to GND.



Step 3 — Upload the Code

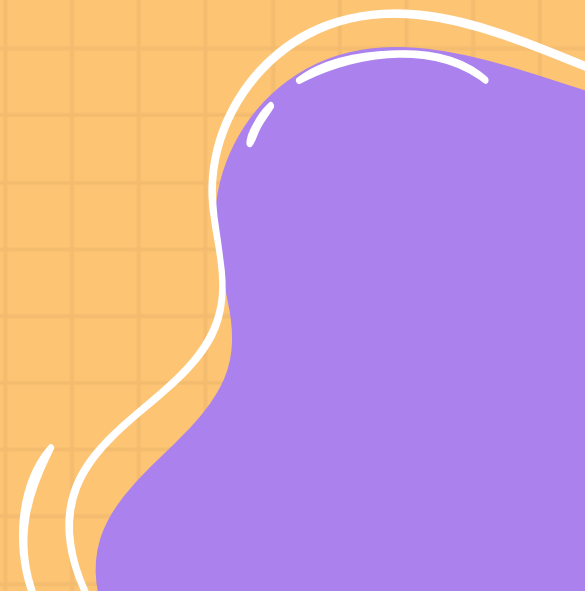
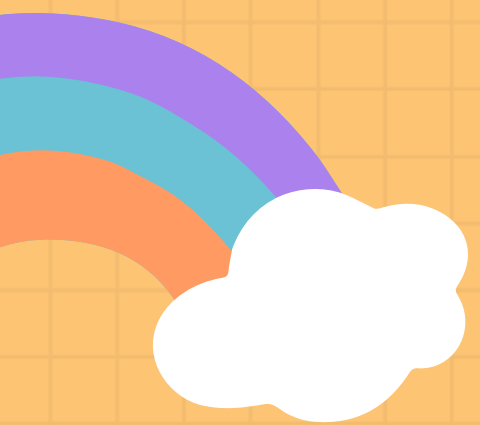


Programming Your RFID Access System

Connect your Arduino to the computer using a USB cable and open the Arduino IDE software.

Upload the code that checks scanned RFID cards against the list of allowed card IDs.

The code controls the LEDs and buzzer based on whether the card is accepted or rejected.



Sample Code Snippet

Simple RFID Card Reading Code

The code reads the unique card ID when you scan your RFID card near the reader.

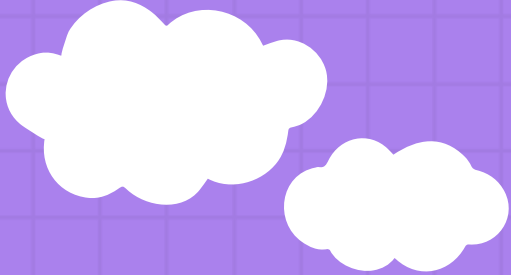
It compares the scanned ID to a list of allowed IDs – if it matches, access is granted!

Step 4 — Test Your System

Time to See Your Project in Action!

Scan your RFID cards near the reader to see if the door unlocks – watch for the green LED to light up and listen for the buzzer beep!

Troubleshoot common issues: check all wire connections are firm, verify card IDs match the code, and look for any error messages in the Arduino IDE.



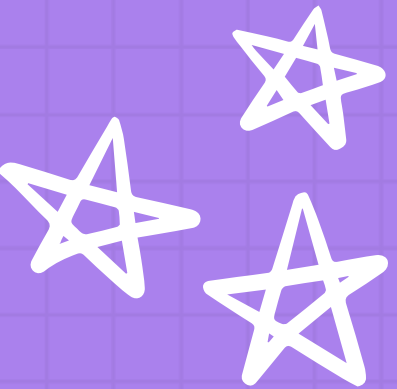
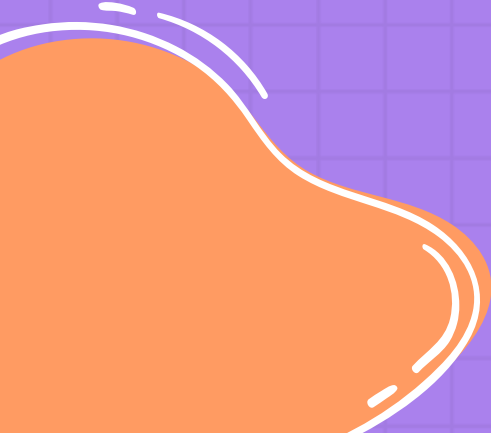
Troubleshooting Tips

Helpful Hints for Common Problems

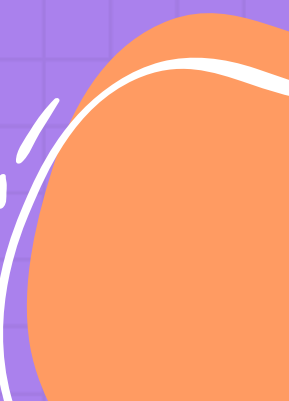
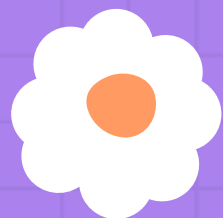
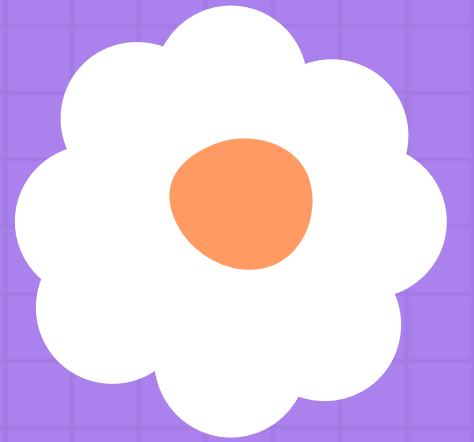
Make sure all wires are connected firmly to the breadboard and Arduino pins.

Check if the correct card IDs are written in your code – even one wrong number won't work!

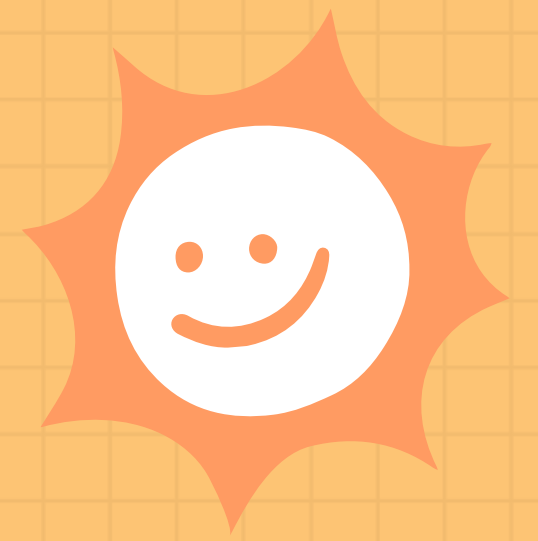
Restart your Arduino by unplugging and plugging it back in, then try again.



Real-World Uses of RFID Access Control



Safety Tips and Best Practices

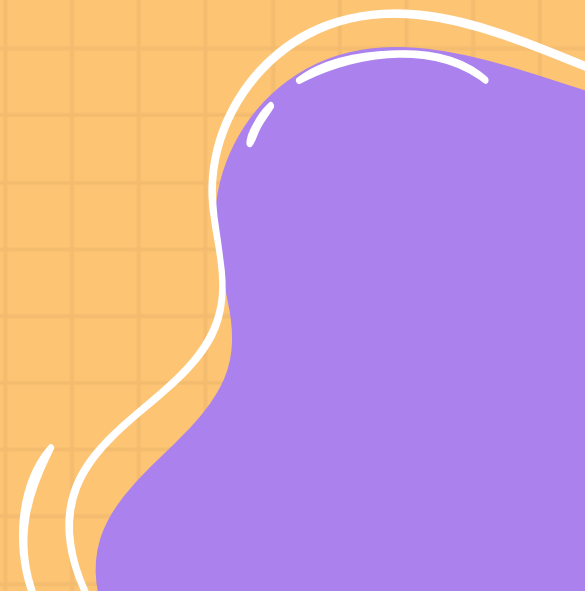
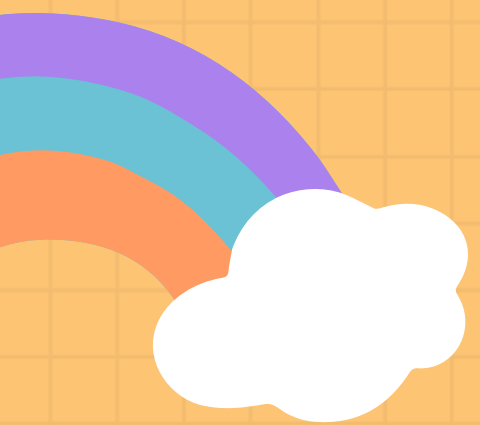


Important Reminders for Working with Electronics

Always unplug your Arduino before changing any wiring or connections.

Avoid touching components when the circuit is powered on to prevent shocks.

Always work with adult supervision when using electronics and tools.





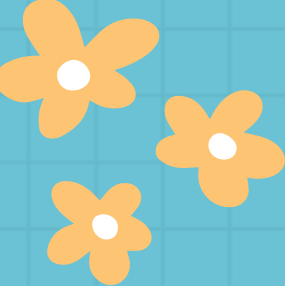
What You Learned Today

key takeaways

How Arduino and RFID work together to create a smart access control system.

Basic electronics wiring skills using breadboards, jumper wires, and components.

Simple programming concepts to read cards and control outputs like LEDs and buzzers.





**Thank You & Keep
Exploring!**

